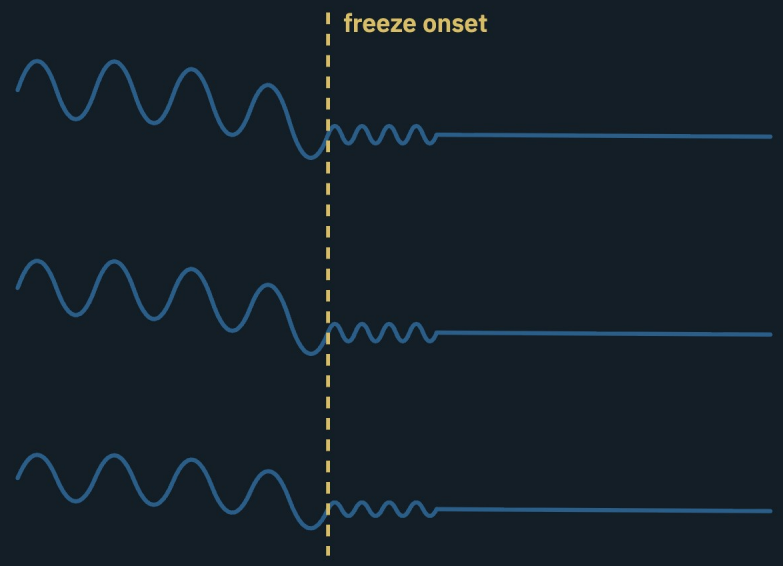
Team Asha · October 2026

A digital twin for freezing of gait

A personalised model of one patient's motor state. It estimates how much that person freezes in daily life — better than the standard clinical questionnaire does in this cohort.



Schematic, not dataset data

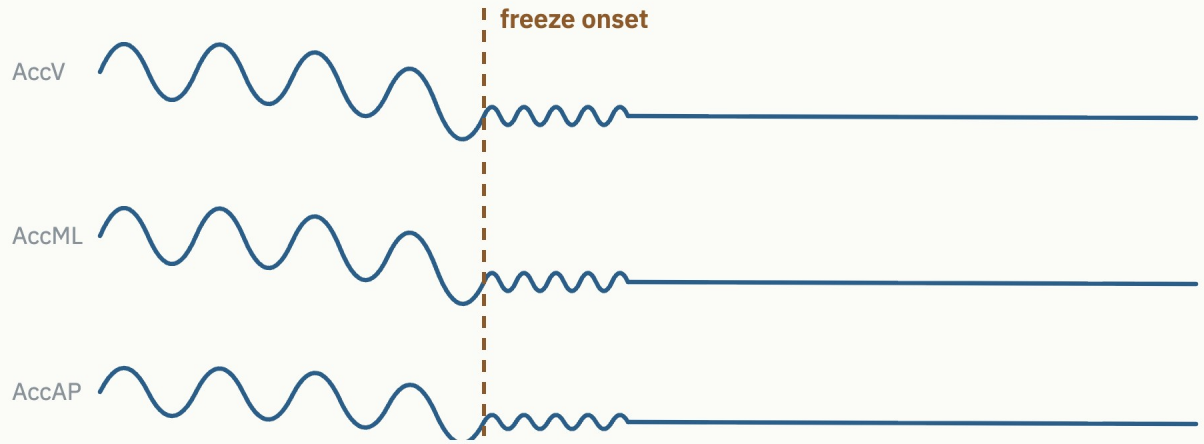
Sir M. Visvesvaraya Institute of Technology, Bengaluru

COHORT
n = 31

LICENCE
MIT

Walking stops, while the intention to walk continues

Strides shorten **then 3–8 Hz trembling**



Schematic, not dataset data. Stride shrinkage and 3–8 Hz trembling before and during a freeze are described in the published literature.

No prevalence figure is quoted: published estimates vary widely by population and assessment method.

Effective forward stepping stops

The person is still trying to walk. The movement does not arrive.

It arrives at the worst moments

Starting to walk, turning, approaching a destination, narrow spaces, or while doing a second task.

> ### A leading cause of falls
>
> Associated with impaired mobility, functional dependency and reduced quality of life. More common with longer disease duration and greater severity.

There is no objective measure of how often this person freezes

WHAT THE CLINICIAN HAS TODAY

Direct observation

Twenty minutes, every three to six months, usually ON medication in a wide corridor — where freezing is least likely to appear.

Patient and carer report

Subject to recall. Arrives as “a few times a week, I think” — an impression, not a quantity you can track.

The NFOGQ questionnaire

Validated, but a single retrospective self-report number, insensitive to change over the intervals that matter.

WHAT THE GAP COSTS — DECISIONS BURDEN FEEDS

01

Medication titration

Is the dose and timing adequate, or is freezing breaking through at end-of-dose?

02

Whether to escalate

Adjunct medication, physiotherapy and cueing referral, or a DBS discussion.

03

Fall-risk stratification

Which drives home modification, gait aids and carer education.

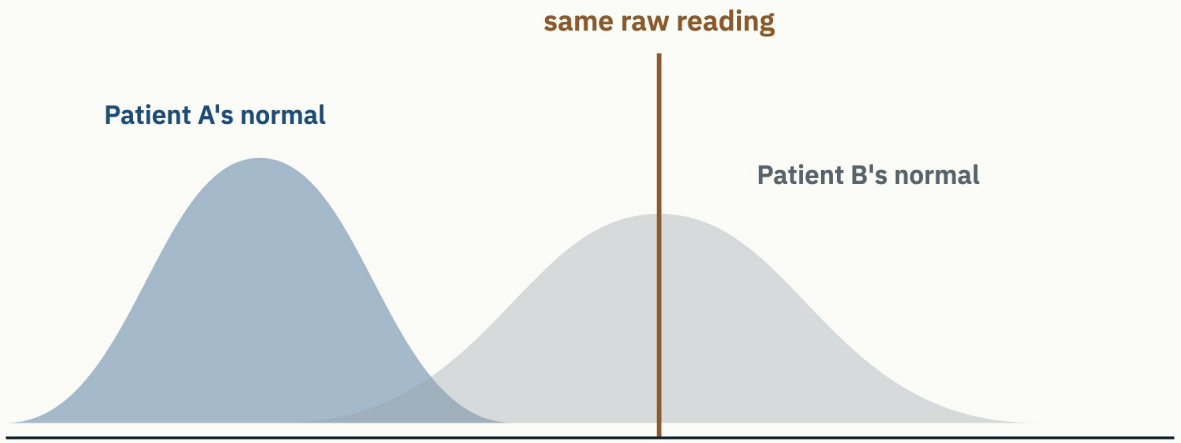
04

Did it work?

After any of the above — whether the thing you did actually changed anything.

Every one of those decisions currently rests on a questionnaire score and an anecdote.

Each window is judged against this patient's own normal



For this identical reading, Patient A is far outside their normal and Patient B is exactly at theirs. A classifier that maps a window to a label cannot tell these two situations apart.

Deviation(t) = ObservedState(t) – PersonalBaseline

1. A week of **unlabelled** home recording sets the baseline, so it cannot contain the freeze labels.
2. Each new 4-second window is compared against that baseline — deviation, not raw movement.
3. Deviation plus the clinical profile produce the two readouts.

> **Why this isn't circular.** Personalisation is learned from data containing none of the answers, so it cannot leak the prediction target.
>
> The twin's state is what makes the comparison possible; whether it improves prediction is an empirical question, and we tested it four ways. All four were null.

Four cohorts, and the 31 where all three streams meet

65 **daily ∩ clinical**
Every daily subject has a clinical record

36 **daily ∩ defog**
Also in the labelled lab cohort

31 **Exact Subject × Visit match**
Clinical, defog and daily on the same person at the same visit. The primary result rests here.

COHORT	SUBJECTS	FILES	ROLE
tdcsfog	62	833	Lab, 128 Hz, m/s². Not used in the reported models — no tdcsfog subject has a personal baseline
defog	45	91	Lab, 100 Hz, g. Valid/Task masks. Evaluation
notype	—	46	Generic event labels. Optional augmentation
daily	65	—	**~1 week continuous, UNLABELLED. Personal baseline**

Zero subjects span tdcsfog and the paired clinical cohort. Other datasets had no patient-level link, so we did not fuse them — merging would invent patients.

Harmonised to 100 Hz and to g. **Labels are index-mapped, never resampled.** Agreement 97.0%, 74 discrepancies, all accounted for. Six-fold subject-level GroupKFold.

Four state layers, two readouts

Static clinical

Demographics · years since diagnosis · UPDRS-III on and off medication · NFOGQ · medication state

Personal baseline — 11 dimensions

Mean of ten movement features plus activity fraction, from the unlabelled week at home

Dynamic

Tri-axial lower-back accelerometry · 4-second windows · 100 Hz

Longitudinal — ordinal only

Visit 1 precedes Visit 2. The elapsed interval is undocumented in this dataset, so no duration is ever attached to it.

READOUT 1 — PRIMARY · ARM A (ACCELEROMETRY ONLY) **Works**

Freezing burden

How much this person freezes in daily life. Tracks observed freezing better than the standard questionnaire.

READOUT 2 — SECONDARY · ARM C (+ BASELINE + CLINICAL STATE) **Weak**

Short-horizon episode risk

Probability that a freeze begins within h seconds. Reported at each h from 0.5 to 5 s, never as one pooled number.

Burden is the spine of this submission. It comes from the accelerometry-only arm: the personalisation branches are the twin's design rationale, and at n = 36 our ablations show they did not improve either readout. We report that rather than assume the architecture earns its keep.

The movement estimate outperforms the questionnaire in this cohort

TWIN BURDEN ESTIMATE

ρ 0.514

p = 0.0031 · Spearman
Pearson r = 0.645, p = 9×10^{-5}

NFOGQ QUESTIONNAIRE ALONE

r 0.353

p = 0.051
Not significant at n = 31

SURVIVES BOTH CONTROLS

0.658
holding activity level

0.609
holding the NFOGQ

A ranking instrument, not a measuring one

It answers **which of my patients freeze most, and is this one getting worse.** It cannot answer *this patient froze twenty-three times last week.* Triage and change-detection, not quantification.

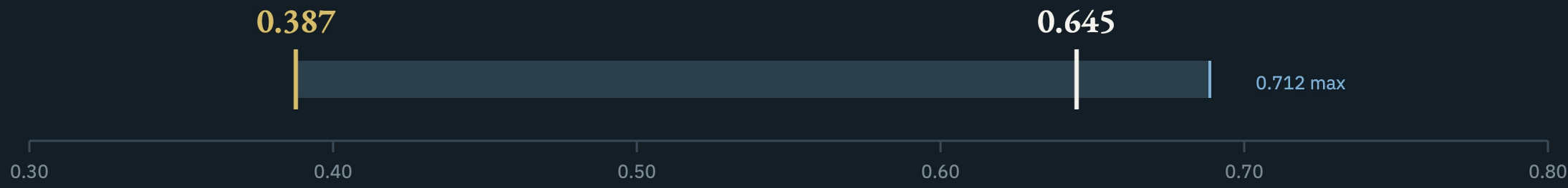
Where they disagree is itself information

The questionnaire captures **perceived** burden; the twin captures **observed** movement. High self-report with a low estimate may be anticipatory avoidance; the reverse suggests under-reporting or adaptation.

All figures from twin_data.json · compared against held-out observed freezing rate

Pearson is leverage-sensitive. Here is exactly how much.

Remove one patient, refit, measure. Repeat for all 31. The relationship holds in **every** refit — but how strong it looks depends noticeably on who is in the sample.



Pearson r across all 31 leave-one-out refits. The band is the full jackknife range; worst-case p = 0.035.

ALL 31 PATIENTS

0.645

$p = 9 \times 10^{-5}$

DROP THE HIGHEST-BURDEN PATIENT

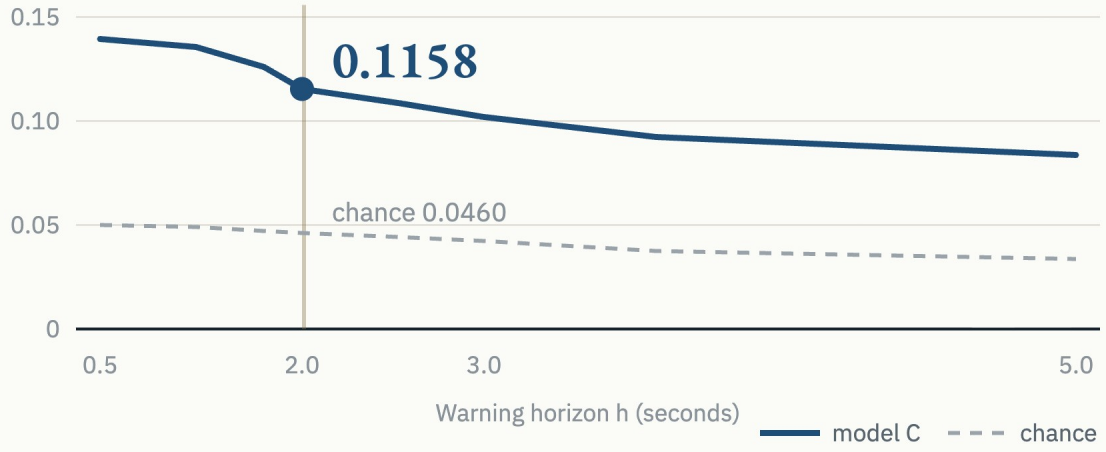
0.387

$p = 0.035$

That is what n = 31 buys you. Which is why **Spearman ρ = 0.514 is the headline**, and why we report a range rather than a point estimate.

Stated here rather than left to be found. A jury that discovers this independently discounts everything else.

Two and a half times chance, decaying with lead time



We never report accuracy. At ~4.6% prevalence, predicting "never freezes" is ~95% accurate and useless. SD across seeds at 2.0 s is 0.0155.

FOUR NULLS, REPORTED AS NULLS

Clinical fusion (C vs A) won 50 of 144	−0.0023
Personal baseline (B vs A) won 46 of 144	−0.0120
Per-patient threshold won 11 of 20	−0.0005
LOSO shrinkage won 8 of 20	−0.0105

WHAT IT COSTS THE PATIENT — TIME ALARMING

10.6%	**19.5%**	**33.3%**
at 30% recall · prec 0.127	at 50% recall · prec 0.114	at 70% recall · prec 0.094

Seed spread for arm C is **0.0295** — an order of magnitude larger than the **0.0023** effect of adding clinical data. Even if fusion helped, we could not tell at this sample size. Model ordering: **A ≈ C > B**.

What we do not claim

Does not diagnose Parkinson's disease, and is not clinically validated, approved or trial-ready.

Does not prevent falls, and has not been shown to improve any clinical outcome.

Concurrent estimate, not a forecast — it describes the week just recorded, not the month ahead.

Ground truth is lab-derived, not home-derived — the home cohort carries no labels.

The dashboard state figure is a schematic indicator keyed to risk, never reconstructed limb motion.

Does not merge patients across datasets, or treat synthetic data as clinical evidence.

A two-second warning helps the patient in the moment. A burden estimate helps the clinician make a decision that shapes the next six months.

One defensible finding, four honest nulls, and a specification of what data would be needed to go further.

Start with docs/04_evidence_matrix.md — every claim is labelled with its evidence class and source.

Team Asha · Sir MVIT, Bengaluru · Happiest Health Digital Twin Challenge 2026

GITHUB	[repository URL]
LICENCE	MIT
TEAM	[members and roles]
CONTACT	[contact email]